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**END OF SEMESTER PROJECT
CLOUD COST AND SECURITY ANOMALY CORRELATION DASHBOARD
BLUE TEAM
CY 376
NETWORK MONITORING, SECURITY AND AUDITING
ASAAH BERNADETTE – FCM.41.018.071.23**

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<https://github.com/stunnerray360-hue/CloudCostSecurityDashboard>

Abstract

Cloud computing has become an important technology for organizations due to its flexibility, scalability, and cost efficiency. However, managing cloud resources introduces challenges such as unexpected cost increases, resource misuse, and security threats. Monitoring cloud expenses and security activities separately may make it difficult to identify relationships between unusual spending patterns and possible security incidents.

This project presents a Cloud Cost & Security Anomaly Correlation Dashboard, a Blue Team security monitoring solution designed to analyze cloud cost behaviour and security events. The system was developed using Python Flask as the application framework, SQLite for data storage, and Chart.js for data visualization. The dashboard collects cloud cost information and security event data, processes the information using a risk correlation algorithm, and produces security findings, risk levels, and anomaly scores.

The developed system monitors cloud spending, displays security alerts, and correlates financial and security activities to identify abnormal behaviour. During testing, the dashboard successfully identified a total cloud cost of \$230.0 and detected three security events consisting of failed login attempts, new user creation, and port scan activity. Based on the correlation analysis, the system generated a MEDIUM risk classification with an anomaly score of 55%, indicating unusual activity that requires further investigation.

The project demonstrates how combining cloud cost monitoring with security event analysis can improve visibility into potential security issues. The solution provides a simple approach for organizations to understand suspicious activities, support security investigations, and improve cloud security monitoring practices.

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1. Introduction

1.1 Background

Cloud computing has become widely used because it provides flexible and scalable resources for organizations. However, managing cloud environments introduces challenges such as unexpected cost increases, resource misuse, and security threats.

Monitoring cloud expenses and security events separately can make it difficult to identify relationships between abnormal spending and possible security incidents. A system that combines both areas can provide better visibility and support faster investigation.

This project focuses on developing a Cloud Cost & Security Anomaly Correlation Dashboard that monitors cloud costs, analyzes security events, and generates risk assessments.

1.2 Problem Statement

Organizations may experience unusual cloud costs due to misconfigurations, excessive resource usage, or unauthorized activities. At the same time, security events such as failed login attempts and suspicious network activities require continuous monitoring.

Without proper correlation between cost and security data, identifying potential risks can be difficult. This project addresses this challenge by combining cloud cost analysis and security event monitoring in one dashboard.

1.3 Aim

The aim of this project is to develop a dashboard that correlates cloud cost data with security events to detect abnormal activities and provide risk findings.

1.4 Objectives

The objectives of the project are to:

- Monitor cloud cost activities.
- Analyze security events.
- Correlate cost and security data.
- Generate risk levels and anomaly scores.
- Provide security findings and recommendations.

1.5 Scope

The project focuses on developing a simulated cloud monitoring system using Flask, SQLite, and Python. It covers cloud cost tracking, security event analysis, risk assessment, and dashboard visualization.

2. Literature and Tooling Review

2.1 Overview

This section discusses the frameworks and technologies that influenced the development of the Cloud Cost & Security Anomaly Correlation Dashboard. The project was guided by cybersecurity concepts including threat behaviour analysis, security monitoring, and cloud security best practices.

2.2 MITRE ATT&CK Framework

The MITRE ATT&CK Framework is a knowledge base that describes common tactics and techniques used by attackers. It helps security teams understand attacker behaviour and improve threat detection.

This project applies MITRE ATT&CK concepts by relating security events such as failed login attempts and port scanning activities to possible attacker actions. This helps in analyzing security alerts and determining potential risks.

2.3 CIS Benchmarks

CIS Benchmarks provide security configuration recommendations for protecting systems and cloud environments. They emphasize secure configurations, access control, and continuous monitoring.

The project follows these principles by monitoring security events and identifying activities that may require investigation

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2.4 SIEM Concepts

Security Information and Event Management (SIEM) systems collect and analyze security events from different sources to detect suspicious activities.

Although this project is not a full SIEM solution, it follows similar principles by collecting security events, analyzing activities, and generating risk findings through correlation

2.5 Technologies Used

Python Flask

Flask was used as the web application framework for developing the dashboard. It manages communication between the database, analysis logic, and user interface.

SQLite

SQLite was used to store cloud cost records and security event data. It provides simple and structured data management for the application.

Chart.js

Chart.js was used to create visual representations of cloud costs and security events through charts, making the results easier to understand.

3. Methodology

3.1 Overview

The project followed a structured approach involving system design, database development, application implementation, and testing. The aim was to develop a dashboard that monitors cloud costs and security events to identify abnormal activities.

3.2 System Architecture

The system consists of four main components:

- **Data Source:** Simulated cloud cost data and security events.
- **Database:** SQLite database used to store project data.
- **Analysis Module:** Python logic used to calculate risk levels.
- **Dashboard:** Flask web application used to display results.

The workflow is:

Cloud Cost Data + Security Events → SQLite Database → Risk Analysis → Dashboard Output

3.3 Development Steps

The project was developed using the following steps:

- Designed the database structure for storing cloud costs and security events.
- Developed the Flask application to connect the database with the dashboard.
- Implemented risk correlation logic to classify activities as Low, Medium, or High risk.
- Added charts and tables for data visualization.
- Tested the dashboard using simulated security and cost data.

3.4 Testing

The system was tested to confirm:

- Data was stored and retrieved correctly.
- Risk calculations worked as expected.
- Charts displayed cloud cost and security information.
- Findings and recommendations were generated successfully.

4. Implementation

4.1 Overview

The implementation phase involved developing the Cloud Cost & Security Anomaly Correlation Dashboard. The system was built using Python Flask, SQLite, HTML, CSS, and JavaScript.

The application collects cloud cost and security event data, processes the information, and displays the results through a web-based dashboard.

4.2 Database Implementation

SQLite was used as the database for storing project data.

Two main tables were created:

Cloud Cost Table

- Stores cloud service names and their costs.
- Example services include EC2, S3, and Database.

Security Events Table

- Stores security event information and severity levels.
- Example events include failed login attempts, new user creation, and port scan detection.

4.3 Flask Application Development

The Flask application (`app.py`) acts as the main controller of the system.

It performs the following tasks:

- Connects to the SQLite database.
- Retrieves cloud cost and security event data.
- Sends data to the dashboard interface.
- Displays risk analysis results.

4.4 Risk Correlation Module

The `correlation.py` file contains the logic used to analyze activities.

The system evaluates:

- Total cloud cost.
- Number of security events.

Based on the results, it assigns a risk level:

- LOW – Normal activity detected.
- MEDIUM – Unusual activity requiring monitoring.
- HIGH – Possible security threat requiring investigation.

4.5 Dashboard Interface

The dashboard was developed using HTML and CSS with Chart.js for visualization.

The interface displays:

- Total cloud cost.
- Number of security alerts.
- Risk level.
- Anomaly score.
- Cloud cost analysis chart.
- Security event chart.
- Security findings and recommendations.

5. Results and Findings

5.1 Overview

This section presents the results obtained after testing the Cloud Cost & Security Anomaly Correlation Dashboard. The system was tested using simulated cloud cost data and security events to evaluate its ability to identify unusual activities.

5.2 Cloud Cost Analysis

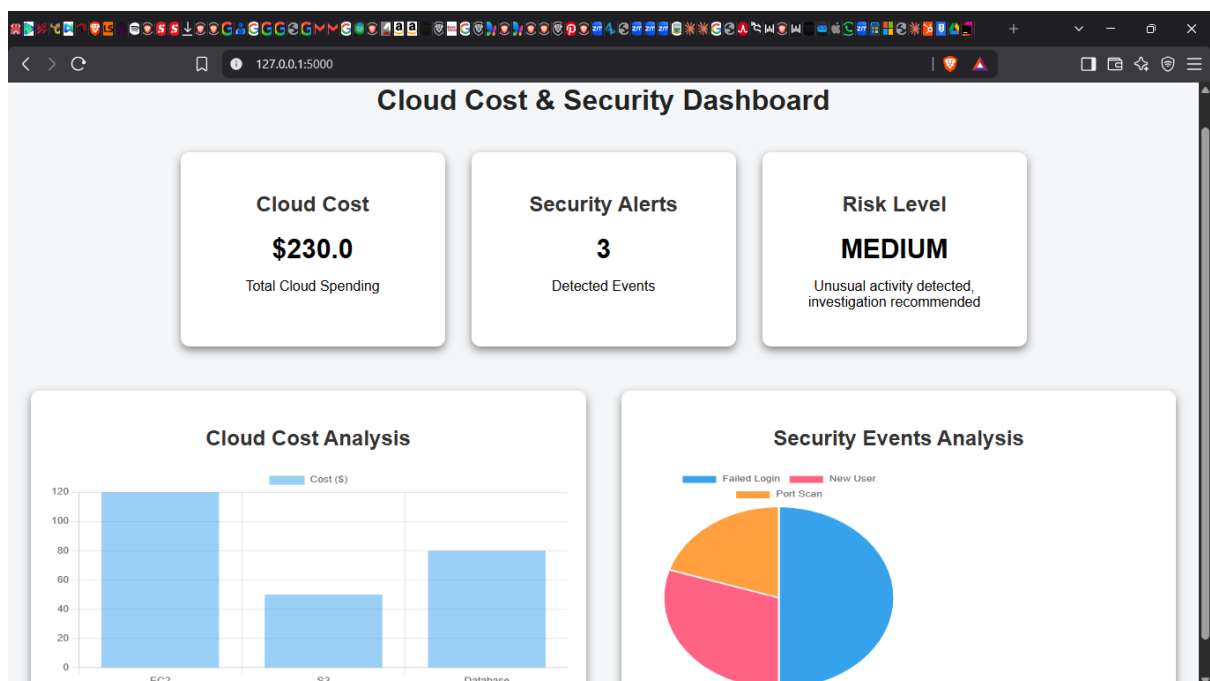
The dashboard successfully displayed cloud resource spending information.

The total cloud cost detected was:

Total Cloud Cost: \$230.0

The cost analysis chart provided a visual representation of spending across different cloud services.

Figure 1: Cloud Cost Analysis Chart



The result shows that the system can collect and display cloud spending information for monitoring purposes.

5.3 Security Event Detection

The system successfully detected and displayed security events stored in the database.

Detected events included:

Security Event	Severity
Failed Login Attempts	High
New User Created	Medium
Port Scan Detected	Low

Figure 2: Security Alert Findings

Security Alert Findings		
Security Event	Severity	Status
Failed Login Attempts	High	Investigation Required
New User Created	Medium	Investigation Required
Port Scan Detected	Low	Investigation Required

These findings show that the dashboard can organize security events and highlight activities that may require investigation.

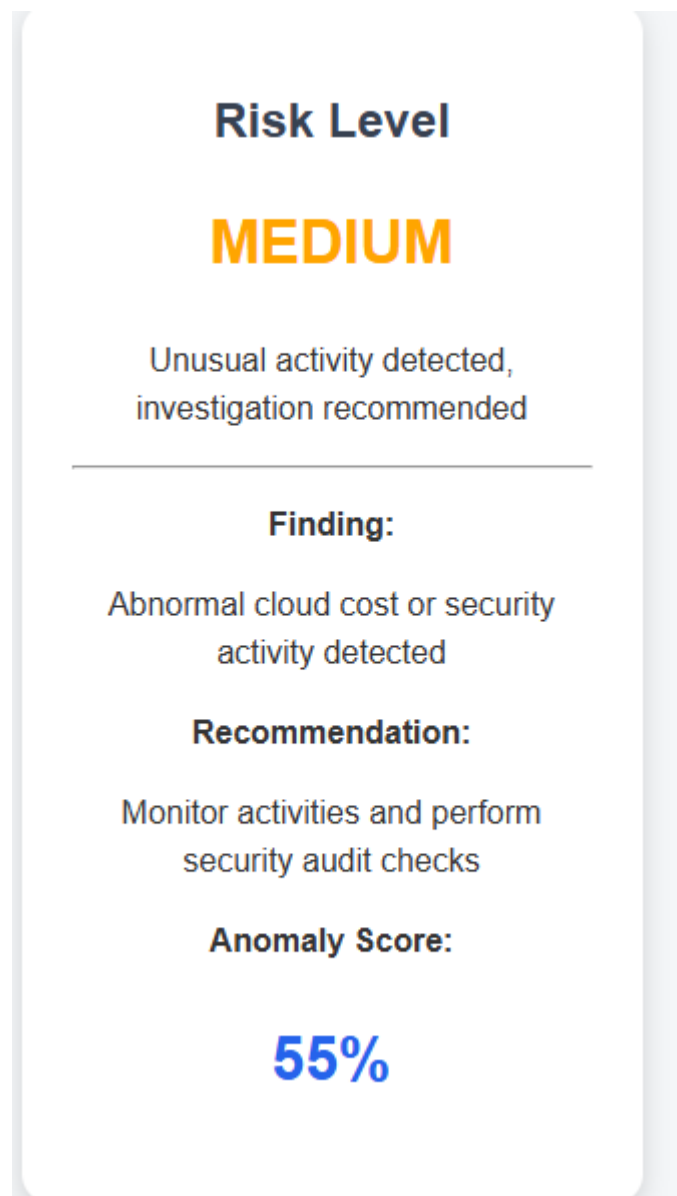
5.4 Risk Assessment Result

After analyzing the cloud cost and security events, the system generated the following result:

Risk Level: MEDIUM

Anomaly Score: 55%

Figure 3: Dashboard Risk Analysis



The MEDIUM risk classification indicates that unusual activity was detected. Further monitoring and security checks are recommended.

6. Analysis and Recommendations

6.1 Analysis of Results

The results obtained from the dashboard show that combining cloud cost information with security events can provide better visibility into unusual activities.

The detected cloud cost of \$230.0 and the presence of security events resulted in a MEDIUM risk classification with an anomaly score of 55%. This suggests that the environment contains activities that should be monitored further.

The failed login attempts may indicate possible unauthorized access attempts, while the port scan activity may represent reconnaissance behaviour. The newly created user account also requires review to ensure that access permissions are properly assigned.

6.2 Recommendations

Based on the findings, the following recommendations are proposed:

- Regularly monitor cloud resource usage to identify unexpected cost increases.
- Review failed login attempts and strengthen authentication controls.
- Apply proper access control and review newly created accounts.
- Monitor network activities for suspicious scanning behaviour.
- Perform regular security audits to identify potential weaknesses.

6.3 Future Improvements

Future improvements to the system may include:

- Integration with real cloud platforms such as AWS or Azure.
- Real-time security event monitoring.
- Use of machine learning models for improved anomaly detection.
- Integration with existing SIEM platforms.

7. Conclusion

The Cloud Cost & Security Anomaly Correlation Dashboard was developed to provide a simple solution for monitoring cloud costs and security events. The project combined cloud spending analysis with security event monitoring to identify unusual activities and generate risk assessments.

The system successfully collected cloud cost data, analyzed security events, and displayed results through a web-based dashboard. The testing process showed that the dashboard could detect security activities, calculate risk levels, and provide recommendations for further investigation.

The project demonstrates the importance of correlating financial and security information when monitoring cloud environments. By combining these areas, organizations can improve visibility, detect suspicious behaviour earlier, and support better security decision-making.

Although the system was developed using simulated data, future improvements can include integration with real cloud platforms, real-time monitoring, and advanced anomaly detection techniques.

8. References

Use one citation style consistently. Example (IEEE style):

[1] MITRE Corporation, "MITRE ATT&CK Framework," Available: <https://attack.mitre.org/>

[2] Center for Internet Security, "CIS Benchmarks," Available: <https://www.cisecurity.org/cis-benchmarks>

[3] Pallets Projects, "Flask Documentation," Available: <https://flask.palletsprojects.com/>

[4] SQLite Documentation, "SQLite Database Engine," Available: <https://www.sqlite.org/docs.html>

[5] Chart.js Documentation, "Chart.js Data Visualization Library," Available: <https://www.chartjs.org/docs/>

9. Appendices

Appendix A: Source Code

```
from flask import Flask, render_template
import sqlite3
from correlation import analyze_risk

app = Flask(__name__)

def get_data():

    connection = sqlite3.connect("database.db")
    cursor = connection.cursor()

    # Total cloud cost
    cursor.execute("SELECT SUM(cost) FROM cloud_cost")
    total_cost = cursor.fetchone()[0]

    if total_cost is None:
        total_cost = 0

    # Security event count
    cursor.execute("SELECT COUNT(*) FROM security_events")
    alerts = cursor.fetchone()[0]

    # Cloud cost chart data
    cursor.execute("SELECT service, cost FROM cloud_cost")
    cost_data = cursor.fetchall()

    services = []
    costs = []

    for item in cost_data:
        services.append(item[0])
        costs.append(item[1])

    # Security events data
    cursor.execute("SELECT event, severity FROM security_events")
    security_data = cursor.fetchall()

    connection.close()
```

```

    return (
        total_cost,
        alerts,
        services,
        costs,
        security_data
    )

@app.route("/")
def dashboard():

    (
        total_cost,
        alerts,
        services,
        costs,
        security_data

    ) = get_data()

    risk, message, finding, recommendation, anomaly_score = analyze_risk(
        total_cost,
        alerts
    )

    return render_template(
        "dashboard.html",
        cost=total_cost,
        alerts=alerts,
        risk=risk,
        message=message,
        finding=finding,
        recommendation=recommendation,
        anomaly_score=anomaly_score,
        services=services,
        costs=costs,
        security_data=security_data
    )

if __name__ == "__main__":
    app.run(debug=True)

```


Appendix B: Database

```
import sqlite3

connection = sqlite3.connect("database.db")

cursor = connection.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS cloud_cost(
    id INTEGER PRIMARY KEY,
    service TEXT,
    cost REAL
)
""")

cursor.execute("""
CREATE TABLE IF NOT EXISTS security_events(
    id INTEGER PRIMARY KEY,
    event TEXT,
    severity TEXT,
    date TEXT
)
""")

cursor.execute("""
INSERT INTO cloud_cost(service, cost)
VALUES
('EC2', 100),
('S3', 50),
('Database', 80)
""")

cursor.execute("""
INSERT INTO security_events(event, severity, date)
VALUES
('Failed Login Attempts', 'High', '15/07/2026'),
('New User Created', 'Medium', '20/07/2026'),
('Port Scan Detected', 'Low', '25/07/2026')
""")

connection.commit()

connection.close()
```

